PROJECT PLANNING, DESIGN & DOCUMENTATION

Agile Sprint Planning

BookMaster – Digital Book Exchange Platform

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1. Sprint Planning Overview

Agile Sprint Planning divides the BookMaster project into smaller, manageable development iterations called **Sprints**.

Each Sprint has:

- A defined goal
- A set of user stories
- Development tasks
- Acceptance criteria
- A potentially usable increment at the end

The BookMaster project will be developed incrementally, with each Sprint building upon the previous one.

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2. Sprint Structure

The project will initially be divided into the following major areas:

Sprint	Focus
Sprint 1	Project setup, database integration and initial Web APIs
Sprint 2	Book catalogue and personal library
Sprint 3	Exchange listings
Sprint 4	Exchange requests and ownership transfer
Sprint 5	Notifications and exchange history
Sprint 6	Testing, optimization and final integration

The exact scope can be adjusted as development progresses.

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3. Sprint 1 – Project Foundation & Core APIs

Sprint Goal

Set up the BookMaster backend and database and implement the foundational Web APIs required for users, categories, and books.

Sprint 1 should establish the foundation required for the exchange functionality that will be implemented in later Sprints.

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4. Sprint 1 User Stories

US-01 – User Creation

As a user, I want to create my account so that I can use BookMaster.

US-02 – User Profile

As a user, I want to view my profile so that I can see my account information.

US-03 – Category Management

As an administrator, I want to manage book categories so that books can be organized properly.

US-04 – Book Creation

As a user, I want to add a book to my library so that I can maintain my collection.

US-05 – Book Catalogue

As a user, I want to browse available books so that I can discover books on the platform.

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5. Sprint 1 Development Tasks

Task 1 – ASP.NET Core Project Setup

- Create ASP.NET Core Web API project.
- Configure application settings.
- Configure dependency injection.
- Configure API routing.

- Configure Swagger/OpenAPI.
- Establish project structure.

Task 2 – Database Setup

- Configure PostgreSQL/SQL Server connection.
- Install and configure Entity Framework Core.
- Create the database context.
- Create initial migrations.
- Create database tables based on the approved schema.

Initial entities:

User

Category

Book

Notification

ExchangeListing

ExchangeRequest

History

Task 3 – User API

Implement:

POST /api/users

GET /api/users/{id}

GET /api/users/{id}/books

The APIs should allow users to be created and their basic information and owned books to be retrieved.

Task 4 – Category API

Implement:

GET /api/categories

GET /api/categories/{id}

POST /api/categories

PUT /api/categories/{id}

DELETE /api/categories/{id}

Task 5 – Book API

Implement the initial book APIs:

GET /api/books

GET /api/books/{id}

POST /api/books

PUT /api/books/{id}

DELETE /api/books/{id}

The system must associate each book with:

- An owner
- A category
- A status

Task 6 – Validation and Error Handling

Implement basic validation for:

- Required fields
- Invalid user IDs
- Invalid category IDs
- Invalid book IDs

- Duplicate/invalid data
- Invalid requests

The API should return appropriate HTTP status codes.

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6. Sprint 1 API Deliverables

At the end of Sprint 1, the following APIs should be available:

Users

POST /api/users

GET /api/users/{id}

GET /api/users/{id}/books

Categories

GET /api/categories

GET /api/categories/{id}

POST /api/categories

PUT /api/categories/{id}

DELETE /api/categories/{id}

Books

GET /api/books

GET /api/books/{id}

POST /api/books

PUT /api/books/{id}

DELETE /api/books/{id}

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7. Sprint 1 Acceptance Criteria

Sprint 1 will be considered complete when:

- The ASP.NET Core Web API project runs successfully.

- The database connection is working.
- EF Core migrations successfully create the required tables.
- Users can be created through the API.
- User information can be retrieved.
- Categories can be created and managed.
- Books can be created and associated with users and categories.
- Books can be retrieved and updated.
- Invalid requests return appropriate errors.
- APIs are documented and testable through Swagger.
- The initial database schema is successfully integrated with the application.

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8. Sprint 1 Deliverable

At the end of Sprint 1, BookMaster should have a working backend foundation:

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        ASP.NET Core Web API
              |
      ┌───────────┼───────────┐
      ▼       ▼       ▼
    Users   Categories   Books
      |       |       |
      └───────────┼───────────┘
              ▼
          Database
```